

Electric circuits virtual laboratories answers

Student self-check booklet

BEFORE YOU CHECK

Complete the investigation first. Then compare your measurements, calculations and explanations with these answers.

Meter readings are approximate. Small differences are acceptable if you used your own readings consistently and included the correct units.

LAB 1 IS CURRENT USED UP

Answer: Disagree with Mia. In one complete series loop, the current is approximately the same at every position.

Current probe position	Expected current I / A
A	about 0.60
B	about 0.60
C	about 0.60

Answer: With the loop broken, the steady current is 0 A. Reconnecting the wire restores the current to about 0.60 A.

LAB 1 EXPLAIN THE FLOW

Answer: Electrons move from the negative terminal towards the positive terminal through the external circuit.

Answer: Conventional current points from the positive terminal towards the negative terminal.

Answer: Changing the display does not change the size of the measured current.

Answer: Charge continues around the loop. The resistor transfers energy; it does not use up charge or current.

Lab 1 Current and charge

Answers for the battery settings and $Q = I \times t$

MEASURED CURRENT

Battery setting / V	Expected current I / A
3	0.30
6	0.60
9	0.90

Answer: When the battery setting increases while resistance stays at 10Ω , the current increases.

CALCULATED CHARGE

Battery / V	Current I / A	Q in 10 s / C	Q in 20 s / C
3	0.30	3.0	6.0
6	0.60	6.0	12
9	0.90	9.0	18

Example working

For 0.60 A during 20 s:

$$Q = I \times t = 0.60 \times 20 = 12 \text{ C}$$

Answer: At the same current, doubling the time makes the charge passing the point twice as large.

Lab 2 Potential difference and energy

Answers for one resistor and $E = V \times Q$

PREDICTION AND MEASUREMENTS

Answer: The 9 V setting transfers more energy to the same 2 C of charge because each coulomb transfers more energy.

Battery setting / V	Battery p.d. / V	Resistor p.d. / V
3	about 3	about 3
6	about 6	about 6
9	about 9	about 9

Answer: In this one-resistor circuit, the resistor p.d. is approximately equal to the battery p.d.

Answer: Both probes on the same ideal wire: about 0 V. Probes at opposite ends of the resistor at the 6 V setting: about 6 V.

CALCULATED ENERGY

Resistor p.d. V / V	Charge Q / C	Energy E / J
3	2	6
6	2	12
9	2	18

Example working $E = V \times Q = 6 \times 2 = 12 \text{ J}$

Answer: A p.d. of 9 V means that 9 J is transferred for every 1 C of charge.

Lab 2 Two resistors and circuit comparison

Answers for energy sharing and equal-current circuits

TWO EQUAL RESISTORS IN SERIES

Answer: Prediction: R1 is about 3 V and R2 is about 3 V.

Measurement	Expected result
Battery p.d.	about 6 V
R1 p.d.	about 3 V
R2 p.d.	about 3 V
Current before R1	about 0.30 A
Current after R1	about 0.30 A

Answer: The current before and after R1 is approximately the same. The p.d. across R1 plus the p.d. across R2 is approximately the battery p.d.

ENERGY ACCOUNT FOR 2 C

Place	p.d. / V	Charge / C	Energy / J
Battery supplies	6	2	12
R1 transfers	3	2	6
R2 transfers	3	2	6

Answer: The current stays the same because charge is conserved. Each resistor transfers energy from the same moving charge.

COMPARE TWO CIRCUITS

Circuit	Current / A	Resistor p.d. / V	Energy per C
A 6 V and 10 Ω	about 0.60	about 6	about 6 J per C
B 12 V and 20 Ω	about 0.60	about 12	about 12 J per C

Answer: The evidence shows that the same current can occur with different potential differences.

Lab 3 Resistance

Answers for $V = IR$ and the graph

FIXED 10 Ω RESISTOR

Answer: If p.d. doubles at fixed resistance, current doubles.

Resistor p.d. V / V	Current I / A	$R = V \div I / \Omega$
2	0.20	10
4	0.40	10
6	0.60	10
8	0.80	10
10	1.00	10
12	1.20	10

Answer: The graph is approximately a straight line through the origin. It shows that V is directly proportional to I when resistance stays constant.

CHANGE THE RESISTANCE AT 6 V

Resistance R / Ω	Resistor p.d. V / V	Current I / A
5	about 6	1.20
10	about 6	0.60
20	about 6	0.30

Answer: If resistance doubles at the same p.d., current halves. As resistance increases, current decreases.

CORE CHECK

- 0.40 A means 0.40 C passes a point each second.
- 8 V means 8 J is transferred for each coulomb.
- $V = IR = 0.20 \times 15 = 3.0$ V.
- Energy is transferred to the surroundings by a resistor.

Challenge answers

Longer explanations and the circuit design

THINK LIKE A PHYSICIST

- Correct the current misconception

Current readings before and after the resistor are approximately equal. Charge continues around the circuit. The resistor transfers energy; it does not use up current.

- Same current with different potential differences

The same current means the same charge passes each second. A resistor with 6 V across it transfers 6 J per coulomb. A resistor with 12 V across it transfers 12 J per coulomb, which is twice as much.

- Evidence from the graph

The points form a straight line through or close to the origin. When current doubles, p.d. doubles. The ratio $V \div I$ remains approximately constant.

DESIGN A CIRCUIT

Find	Working	Answer
Current I	$I = Q \div t = 3.0 \div 5.0$	0.60 A
Potential difference V	$V = E \div Q = 18 \div 3.0$	6.0 V
Resistance R	$R = V \div I = 6.0 \div 0.60$	10 Ω

Answer: Build the circuit using a 6 V battery and a 10 Ω resistor.

Answer: Expected current: about 0.60 A. Expected resistor p.d.: about 6 V.

Answer: Check: $Q = I \times t = 0.60 \times 5.0 = 3.0$ C.

Answer: Check: $E = V \times Q = 6.0 \times 3.0 = 18$ J.